

Lec2

Semantic Tags

Semantic tags in HTML are tags that clearly describe the meaning and purpose of the content they contain. They make web pages easier to read for developers, browsers, and search engines.

Tag	Purpose
<header>	Defines the header section of a page or section
<nav>	Represents navigation links
<main>	Specifies the main content of the document
<section>	Groups related content together
<article>	Represents independent content like blogs or news articles
<aside>	Contains side content such as sidebars or ads
<footer>	Defines the footer section
<figure>	Groups media content like images or diagrams
<figcaption>	Adds a caption for a figure
<details>	Creates expandable details content
<summary>	Defines the heading for <details>

Block Elements

Block elements take the full width available and always start on a new line.

Examples

`<div>`

`<p>`

`<h1>` to `<h6>`

`<section>`

`<article>`

Inline Elements

Inline elements only take as much width as needed and do not start on a new line.

Examples

`<span>`

`<a>`

`<strong>`

`<em>`

`<img>`

Difference between block and inline element

Block Elements	Inline Elements
Start on a new line	Do not start on a new line
Take full width available	Take only required width
Can contain inline and block elements	Usually contain text or other inline elements
Used for page structure	Used for styling/content within a line

Basic Tags

Lists

Table

Forms

Lists

HTML offers three ways for specifying lists: ordered lists, unordered lists, and description lists. Ordered lists use ordinal sequences to indicate the order of list elements, unordered lists use a defined symbol such as a bullet to list elements in no designated order, and description lists use indents to list elements with their children.

Ordered List

An ordered list can be created with the `<ol>` tag and each list item can be created with the `<li>` tag

```
<ol>  
  <li>Item</li>  
  <li>Another Item</li>  
  <li>Yet Another Item</li>  
</ol>
```

Changing the type of numeral

1 Default value -Decimal numbers 1,2,3,4

a Alphabetically ordered (lowercase) a,b,c,d

A Alphabetically ordered (uppercase) A,B,C,D

i Roman Numerals (lowercase) i,ii,iii,iv

I Roman Numerals (uppercase) I,II,III,IV

Unordered List

An unordered list can be created with the `<ul>` tag and each list item can be created with the `<li>` tag.

```
<ul>  
  <li>Item</li>  
  <li>Another Item</li>  
  <li>Yet Another Item</li>  
</ul>
```

Nested List

```
<ul>  
  <li>item 1</li>  
  <li>item 2  
    <ul>  
      <li>sub-item 2.1</li>  
      <li>sub-item 2.2</li>  
    </ul>  
  </li>  
  <li>item 3</li>  
</ul>
```

Description List

A description list (or definition list, as it was called before HTML5) can be created with the dl element. It consists of name-value groups, where the name is given in the dt element, and the value is given in the dd element.

```
<dl>
  <dt>name 1</dt>
  <dd>value for 1</dd>
  <dt>name 2</dt>
  <dd>value for 2</dd>
</dl>
```

Tables

The HTML `<table>` element allows web authors to display tabular data (such as text, images, links, other tables, etc.) in a two dimensional table with rows and columns of cells.

Simple Table

```
<table>
  <tr>
    <th>Heading 1/Column 1</th>
    <th>Heading 2/Column 2</th>
  </tr>
  <tr>
    <td>Row 1 Data Column 1</td>
    <td>Row 1 Data Column 2</td>
  </tr>
  <tr>
    <td>Row 2 Data Column 1</td>
    <td>Row 2 Data Column 2</td>
  </tr>
</table>
```

Spanning columns or rows

Table cells can span multiple columns or rows using the colspan and rowspan attributes. These Attributes Can Be Applied To <th> and <td> elements.

row 1 col 1	row 1 col 2	row 1 col 3
This second row spans all three columns		
This cell spans two rows	row 3 col 2	row 3 col 3
	row 4 col 2	row 4 col 3


```
<table>
  <tr>
    <td>row 1 col 1</td>
    <td>row 1 col 2</td>
    <td>row 1 col 3</td>
  </tr>
  <tr>
    <td colspan="3">This second row spans all three columns</td>
  </tr>
  <tr>
    <td rowspan="2">This cell spans two rows</td>
    <td>row 3 col 2</td>
    <td>row 3 col 3</td>
  </tr>
  <tr>
    <td>row 4 col 2</td>
    <td>row 4 col 3</td>
  </tr>
</table>
```

Forms

In order to group input elements and submit data, HTML uses a form element to encapsulate input and submission elements. These forms handle sending the data in the specified method to a page handled by a server or handler.

Example:

```
<form action="action.php" method="post" target="_blank" accept-charset="UTF-8"  
enctype="application/x-www-form-urlencoded" autocomplete="off" novalidate>
```

```
<input type="text" name="lastname" value="Mouse">
```

Attribute	Description
<code>accept-charset</code>	Specifies the character encodings to be used for form submission.
<code>action</code>	Specifies where to send the form data when the form is submitted.
<code>autocomplete</code>	Specifies whether the form should have autocomplete on or off.
<code>enctype</code>	Specifies how the form data should be encoded when submitting it to the server (used with <code>method="post"</code>).
<code>method</code>	Specifies the HTTP method used when sending form data (GET or POST).
<code>name</code>	Specifies the name of the form.
<code>novalidate</code>	Specifies that the form should not be validated when submitted.
<code>target</code>	Specifies where to display the response received after submitting the form.

Target attribute in form tag

The target attribute specifies a name or a keyword that indicates where to display the response that is received after submitting the form.

The target attribute defines a name of, or keyword for, a browsing context (e.g. tab, window, or inline frame)

From Tag with a target attribute:

```
<form target="_blank">
```

Value	Description
<code>_blank</code>	The response is displayed in a new window or tab.
<code>_self</code>	The response is displayed in the same frame (default behavior).
<code>_parent</code>	The response is displayed in the parent frame.
<code>_top</code>	The response is displayed in the full body of the window.
<code>framename</code>	The response is displayed in a named iframe.

Example

```
<body>
<h2>Student Form</h2>
<form action="submit.php" method="post">
Name:
<input type="text" name="name"><br><br>
Email:
<input type="email" name="email"><br><br>
Password:
<input type="password" name="password"><br><br>
<input type="submit" value="Submit">
</form>
</body>
</html>
```

Homework

	1	2	3		4	6		
Mon	C++	HTML	CSS	BREAK	JAVA	C++		
Tue	Hindi	Webdev			C++	HTML	CSS	OS
wed		C++			Django			
Thu		Java			English	Go	Networks	
Fri		Data Science			C++		Go	Python

Task

Create a semantic webpage with table and form elements